

CREATING A MODEL BY AMAZON SAGEMAKER AI

Name - RRAMANDEIP SINGH
Student ID - N10010960

CONTENTS

Introduction

Dataset Description

Data Exploration

Data Preprocessing

Feature Selection

Model Development

Model Evaluation

Hyperparameter Tuning

Data Visualization

Cloud Deployment using AWS SageMaker

Results and Predictions

Conclusion

Introduction

In this project, the main aim is to create a machine learning algorithm that will be able to categorise Netflix movies as MOVIE and SHOW.

As the number of streaming platforms grows at an alarming rate, the importance of content classification for recommendation algorithms grows as well.

The purpose of this project is to develop an efficient and scalable classifier using structured features such as runtime, release date, IMDb rating, TMDb rating, and popularity.

Dataset Description

The dataset that has been utilized for this project is the Netflix titles together with their features. These features include:

- Runtime - The length of the movie
- Year - Year of release
- IMDb Score - Rating on IMDb
- TMDb Score - Rating on TMDb
- TMDb Popularity - Popular index

Target variable being considered is 'type', which refers to MOVIE or SHOW.

Data Exploration

Preliminary analysis was carried out using the panda's library commands like head(), info(), describe(), and shape.

These operations were crucial in getting an insight into the structure of our data, the presence of missing values, and the analysis of distributions.

It became evident from exploratory analysis that some columns had missing values.

Data Preprocessing

Data preprocessing is an essential phase in machine learning. In this assignment:

- The missing data in the relevant numeric columns was eliminated through the use of dropna()
- The relevant variables were chosen for learning
- The target value was numerically encoded

- The data was ready for learning

This process guaranteed that the dataset was ready for machine learning purposes.

Feature Selection

The above-listed attributes have been chosen due to their significance:

- runtime
- release_year
- imdb_score
- tmdb_score
- tmdb_popularity

The above-listed attributes offer valuable information about content-related properties that enhance model accuracy.

Model Development

For the present project, a Random Forest Classifier was employed because of its resilience and effectiveness in dealing with complicated associations within datasets.

Data was split between train and test data sets at a ratio of 80:20.

The machine learning model was then trained on the training data set and tested on the test data set.

Model Evaluation

It had an accuracy rate of about 93.7%.

Other evaluation methods include the following:

- Confusion Matrix
- Precision and Recall
- F1 Score

All these prove that the model can easily differentiate between movies and TV series.

Hyperparameter Tuning

For optimisation of the model, hyperparameter tuning was conducted via GridSearchCV.

Tuning of parameters like estimators and maximum depth was done.

There was improvement in consistency while still retaining accuracy.

Data Visualization

The use of visualization methods helped in gaining more insights from the data set:

- Correlation Heatmap for feature correlation
- Confusion Matrix Heatmap for predicting performance
- Distribution plot for IMDb scores
- Scatterplot for Runtime against Ratings

Cloud Deployment using AWS SageMaker

Deployment was performed by deploying the trained model on AWS SageMaker in real-time.

Deployment steps were as follows:

- Dataset upload on Amazon S3
- Training job creation via SKLearn estimator
- Execution of the training script (train.py)
- Deployment of the model as an endpoint
- Prediction tests through API calls

Results and Predictions

The trained model effectively produced results with the help of varied inputs.

Illustration:

Input: [120, 2018, 8.0, 7.5, 80] → Result: MOVIE

The model gave sensible results each time on the basis of the inputs.

Conclusion

Out of all the models that have been evaluated in this experiment, it can be seen that the Random Forest model performed excellently, scoring a high accuracy score of around 93.7%, which proves the effectiveness of the algorithm to deal with classification tasks and find patterns within the dataset.

In addition to that, using the AWS SageMaker platform has helped to streamline the development process of the machine learning model and demonstrate the practicality of cloud-based machine learning services.

In conclusion, this experiment demonstrates that by combining machine learning algorithms with appropriate data processing methods and leveraging cloud computing technology, one can design effective and scalable solutions to address classification problems.